Theoretical Bounds on χ_ν in Observer-Patch Holography

Coherent-Matter Susceptibility and the Construction Window

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Abstract

The OPH coherent-matter scalar branch contains a susceptibility coefficient. On the uniform product-thickening branch with slice-wise scalar-reserve unbiasedness, the canonical value is $\chi_\nu = e^{-P/24} = 0.9343006394893864\ldots$, with P the public endpoint scalar. In the engineering chart used here, the same invariant coefficient appears as a very small vertical response.

The estimate turns the branch into a laboratory target. For a $100\,\mathrm{kg\,m^{-2}}$ platform, full weight response asks for coherence contrast near 10^{-8}. Ten percent assist asks for contrast near 10^{-9}. The substrate question is open: a controlled experiment must show whether real materials can produce that scalar contrast.

What This Paper Contributes

This note takes one dark-sector scalar coefficient and turns it into an experimental target. The OPH core supplies the observer carrier, record algebras, quotient observables, and canonical collar scalar channel. The added branch supplies a finite source-generator test for whether coherent matter can source that same channel in a controlled material system.

The result is useful because it separates the invariant coefficient from the engineering chart. The canonical susceptibility is order one on the exact uniform branch, while the laboratory response can look tiny after stored coherent energy and effective capacity are folded into the scalar. The paper therefore gives a narrow construction window: what contrast a platform must produce, what a null result would mean, and which part is a continuation instead of a recovered-core OPH claim.

Contents

1 Introduction

Four layers are separated throughout the note. The recovered OPH core supplies the fixed-cutoff observer carrier, record algebras, quotient-local observables, the Einstein branch, and the canonical scalar source surface. Tier A alone leaves χ_ν unassigned.

Coherent-matter same-channel forcing supplies the finite source theorem. A nondegenerate OPH-coherent material subfederation has a canonical scalar source

$$S_{\text{coh}}^{\text{can}} = \mathbf{1}_{\text{self-read}}\,\mathsf{R}_U\,\mathsf{P}_U\,\mathsf{C}_U,$$

and Scalar Edge-Center Exhaustion forces that source into the same edge-center scalar register priced by the Z_6 collar theorem. The zero branch is the branch where at least one coherent-material receipt vanishes.

The protected-reserve collar theorem fixes the canonical coefficient on the closed collar branch. Granting the dark-sector collar lemmas on that shared register, the canonical susceptibility is the collar survival coefficient:

$$\chi_\nu^{\text{can}} = \lambda_{\text{collar}}.$$

Engineering calculations often use a different chart. They fold stored coherent energy into the scalar. That makes the coefficient look tiny, although the canonical coefficient is order one.

The canonical number is the OPH-invariant number. It uses the per-channel observer-facing scalar after the effective patch capacity has been divided out. On the exact uniform product-thickening branch,

$$\chi_\nu^{\text{can}} = e^{-P/24} = 0.9343006394893864\ldots.$$

With finite-thickness averaging and the protected reserve mean $P/24$, the theorem-level band is

$$\boxed{0.9343006394893864 \le \chi_\nu^{\text{can}} \le 1.}$$

The interval is narrow enough for engineering purposes. The exact branch is at the lower end of this interval.

The engineering chart number is

$$S_{\text{coh}}^{\text{eng}} = N_{\text{coh}} S_{\text{coh}}^{\text{can}}, \qquad N_{\text{coh}} = \varepsilon_{\text{sub}} \frac{u_{\text{stored}}}{u_0}.$$

The coefficient in that chart is smaller by the same factor:

$$\boxed{\chi_\nu^{\text{eng}} = \frac{\chi_\nu^{\text{can}}}{N_{\text{coh}}}.}$$

This gives the engineering band

$$\boxed{\frac{0.9343006394893864}{N_{\text{coh}}} \le \chi_\nu^{\text{eng}} \le \frac{1}{N_{\text{coh}}}.}$$

The translation is simple:

N_{coh}	χ_ν^{eng} range
10^{20}	9.34×10^{-21} to 1.00×10^{-20}
10^{21}	9.34×10^{-22} to 1.00×10^{-21}
10^{22}	9.34×10^{-23} to 1.00×10^{-22}
10^{23}	9.34×10^{-24} to 1.00×10^{-23}
10^{24}	9.34×10^{-25} to 1.00×10^{-24}

The engineering consequence is direct. The coefficient is large enough in the canonical chart. The hard part is the scalar: build a substrate with enough vertical canonical coherence contrast, keep ordinary ambient matter from creating the same scalar accidentally, and measure the force with clean controls. Values around 10^{-22} mean $N_{\text{coh}} \sim 10^{22}$, while the canonical coefficient stays order one.

2 What the Bound Says for Devices

The branch theorem gives the canonical coefficient:

$$0.9343006394893864 \leq \chi_\nu^{\text{can}} \leq 1.$$

On the exact uniform product-thickening branch it is the single number

$$\chi_\nu^{\text{can}} = 0.9343006394893864\ldots.$$

The susceptibility bound belongs to the closed quotient-edge collar branch. It is an order-one coefficient. Zero is excluded for nondegenerate coherent-material sources on that branch.

A room-scale lift or weight-reduction device needs only tiny $\Delta\nu$. For areal load $\Sigma = M/A_\perp$ and target weight fraction f, the required response is

$$\Delta\nu_{\text{req}} = f\,\frac{4\pi G\Sigma}{g} = 8.5525 \times 10^{-9} f\,\Sigma_2, \qquad \Sigma_2 = \frac{\Sigma}{10^2\,\text{kg}\,\text{m}^{-2}}.$$

The forced χ_ν^{can} band converts this into the required canonical coherence contrast:

$$\boxed{8.55 \times 10^{-9} f\,\Sigma_2 \leq \Delta S_{\text{coh,req}}^{\text{can}} \leq 9.15 \times 10^{-9} f\,\Sigma_2.}$$

For devices, the number to build is the coherence contrast. A $100\,\text{kg}\,\text{m}^{-2}$ room-scale platform requires a vertical canonical coherence contrast of roughly 8.6×10^{-9} to 9.2×10^{-9} for full weight compensation. Ten percent compensation requires roughly 8.6×10^{-10} to 9.2×10^{-10}.

For a hoverboard-class footprint the areal load is usually higher. A rider plus board spread over $0.5\,\text{m}^2$ gives $\Sigma \simeq 200$ to $300\,\text{kg}\,\text{m}^{-2}$. Full response then asks for

$$\Delta S_{\text{coh,req}}^{\text{can}} \simeq 1.7 \times 10^{-8} \text{ to } 2.7 \times 10^{-8}.$$

A compact footprint near $600\,\text{kg}\,\text{m}^{-2}$ asks for about

$$5.1 \times 10^{-8} \text{ to } 5.5 \times 10^{-8}.$$

Device-scale case	$\Sigma\,[\mathrm{kg\,m^{-2}}]$	f	$\Delta\nu_{\mathrm{req}}$	$\Delta S^{\mathrm{can}}_{\mathrm{coh,req}}$
light room platform	50	1	4.28×10^{-9}	4.28×10^{-9} to 4.58×10^{-9}
room-scale platform	100	1	8.55×10^{-9}	8.55×10^{-9} to 9.15×10^{-9}
heavy room platform	250	1	2.14×10^{-8}	2.14×10^{-8} to 2.29×10^{-8}
ten percent assist	100	0.1	8.55×10^{-10}	8.55×10^{-10} to 9.15×10^{-10}

Table 1: Room-scale response requirements implied by the branch-theorem canonical band. The table is independent of the engineering chart normalization.

Ten percent assist divides these numbers by ten. The coefficient is in the useful range for hoverboard-class experiments. A practical hoverboard depends on whether a real substrate can produce and hold that vertical scalar contrast under load.

The engineering coefficient is a chart value:

$$\chi_\nu^{\mathrm{eng}} = \frac{\chi_\nu^{\mathrm{can}}}{N_{\mathrm{coh}}}.$$

The small engineering numbers are useful only together with the large engineering scalar. The response depends on the product

$$\chi_\nu^{\mathrm{eng}}\,\Delta S^{\mathrm{eng}}_{\mathrm{coh}} = \chi_\nu^{\mathrm{can}}\,\Delta S^{\mathrm{can}}_{\mathrm{coh}}.$$

For $N_{\mathrm{coh}} = 10^{22}$, the forced engineering band is

$$9.34 \times 10^{-23} \le \chi_\nu^{\mathrm{eng}} \le 1.00 \times 10^{-22}.$$

A $100\,\mathrm{kg\,m^{-2}}$ full-response platform then needs

$$\Delta S^{\mathrm{eng}}_{\mathrm{coh,req}} \simeq 8.6 \times 10^{13} \text{ to } 9.2 \times 10^{13}.$$

It equals 10^{22} times the canonical contrast in Table 1. The large scalar and small coefficient are the same normalization written in different coordinates.

On the closed collar branch, the coefficient is in the useful mathematical range for macroscopic weight-reduction devices. No room-temperature material claim follows from this bound alone. The engineering task is specific: generate a stable vertical $\Delta S^{\mathrm{can}}_{\mathrm{coh}}$ near 10^{-9} for assist and near 10^{-8} to 10^{-7} for full hoverboard-class response, keep ambient ordinary matter from creating the same scalar accidentally, and measure the force with controls that separate the effect from acoustic, thermal, electromagnetic, and mechanical couplings.

3 Support Boundary

The paper uses three support levels.

Tier A is the recovered OPH core: finite patch carriers, mismatch-lowering repair, observer-accessible record algebras, quotient normal forms, checkpoint continuation, controlled BW scaling, fixed-cap generalized-entropy stationarity, and the Jacobson-type Einstein branch. These ingredients are supplied by the consensus, microphysics, synthesis, and compact SM/GR papers [3, 4, 1, 2]. This tier leaves χ_ν unfixed.

Tier B0 is the finite coherent-matter source theorem. It defines a self-reading material subfederation, its quotient-visible scalar-source readout, and a finite edge-center source generator. Under Scalar Edge-Center Exhaustion, every nondegenerate source $S_{\rm coh}^{\rm can}$ is forced into the same quotient-edge scalar register used by the dark-sector collar channel. Tier B0 derives the channel identity; it does not assign the reserve survival coefficient.

Tier B1 is the coherent-matter response law

$$\delta\nu = \chi_\nu^{\rm can} S_{\rm coh}^{\rm can} = \chi_\nu^{\rm eng} S_{\rm coh}^{\rm eng}. \tag{3.1}$$

This tier states linear weak-field response and the two charts without assigning a value for χ_ν.

Tier C is the conditional quotient-edge collar branch. It adds the dark-sector collar lemmas stated in Section 8. On this branch, coherent scalar opportunities are co-registered with the canonical dark-sector scalar repair channel, and $\chi_\nu^{\rm can}$ is forced to equal $\lambda_{\rm collar}$.

The dark-sector paper supplies the settled weak-field anomaly law

$$\rho_A = -\frac{1}{4\pi G}\nabla\cdot[(\nu_{\rm OPH} - 1)\mathbf{g}_b], \tag{3.2}$$

interpreting the anomaly as a transported modular/collar information-defect remainder, with no additional particle species postulated [5].

Here $S_{\rm coh}^{\rm can}$ is the per-channel observer-facing scalar. It enters the dark-sector scalar channel only through the finite source lift in Section 7. The scalar $S_{\rm coh}^{\rm eng}$ is the stored-energy engineering scalar. The small values useful in device estimates are the engineering chart values $\chi_\nu^{\rm eng} = \chi_\nu^{\rm can}/N_{\rm coh}$.

4 Canonical Chain

4.1 Fixed-cutoff carrier

At fixed cutoff, an observer-facing OPH implementation surface is a federated patch carrier

$$\mathfrak{F} = (V, E, \{\mathcal{A}_i\}_{i\in V}, \{\mathcal{I}_e\}_{e\in E}, \{\pi_{i,e}\}, \{\mathcal{R}_i\}, \{\mathcal{U}_i\}). \tag{4.1}$$

Here $\mathcal{A}_i$ are local finite algebras, $\mathcal{I}_e$ are overlap interfaces, $\pi_{i,e}$ are visible restriction maps, $\mathcal{R}_i \subseteq Z(\mathcal{A}_i)$ are record algebras, and $\mathcal{U}_i$ are local update and repair interfaces. Physical claims are made through visible restrictions, record algebras, and quotient-local observables, without choosing hidden microscopic representatives [4].

4.2 Mismatch and repair

The consensus paper gives a nonnegative visible mismatch functional Φ. Accepted repairs lower the relevant mismatch on finite state spaces. Under the declared local-diamond and repair-completeness hypotheses, the terminal observer-facing normal form from a fixed initial physical quotient state is unique and independent of asynchronous update schedule [3].

4.3 Records and checkpoints

For an observer-supporting subfederation U, the microphysics paper uses a checkpoint of the form

$$\mathrm{Chk}_U(t) = \big(\mathcal{R}_U(t), \rho_U^{\mathrm{acc}}(t), \mathfrak{I}_U^{\mathrm{ext}}(t), \nu_{\geq t}, \mathfrak{B}_U(t)\big). \tag{4.2}$$

If two checkpoints agree on the accessible record algebra, accessible state, external interface, boundary-update schedule class, and provenance bundle, they induce the same continuation law on the observer-accessible event algebra. Approximate agreement gives a controlled total-variation bound [4].

4.4 Gravity and dark response

The compact SM/GR paper keeps the recovered core and downstream continuations separate. On the stated controlled scaling branch, the OPH stack gives a Jacobson-type Einstein relation; dark-sector response laws are phenomenological continuations outside that recovered-core tier [2]. The dark paper then models the anomaly by Eq. (3.2).

On Earth, $g_b/a_0 \sim 10^{11}$, so the canonical ν_{OPH} term is pinned near unity. A terrestrial tabletop anomaly attributed only to the canonical OPH dark sector should read null. The χ_ν continuation is tested against that null baseline as a separate effect.

5 The Coherence Scalar

Definition 5.1 (Patch coherence factors and channel count)**.** *Let U be an observer-supporting patch subfederation at cycle t, with observer-accessible record algebra $\mathcal{Z}_U^{\mathrm{rec}}(t)$. Let $Y_U(t)$ be the induced record outcome variable, $X_{\partial U,t+1:t+h}$ the boundary packet over horizon h, and $\Phi_U(t)$ the visible mismatch on the relevant support. Define*

$$R_U(t;h) = \left[1 - d_{\mathrm{TV}}(\mathrm{Law}(Y_U(t+h)), \mathrm{Law}(Y_U(t)))\right] e^{-\delta_{\mathrm{rec}}(t)}, \tag{5.1}$$

$$P_U(t;h) = \frac{I(Y_U(t); X_{\partial U,t+1:t+h})}{H(X_{\partial U,t+1:t+h}) + \epsilon}, \tag{5.2}$$

$$C_U(t) = \left[1 - \frac{\Phi_U(t)}{\Phi_U^{\max} + \epsilon}\right]_+. \tag{5.3}$$

On the screen or hardware branch, set the finite coherent channel count to

$$N_U(t) := \mathrm{Cap}(\mathcal{Z}_U^{\mathrm{rec}}(t)). \tag{5.4}$$

The exponent-free observer core is

$$\mathfrak{c}_U(t;h) = \mathbf{1}_{\text{self-read}}\, R_U(t;h)\, P_U(t;h)\, C_U(t). \tag{5.5}$$

Definition 5.2 (Canonical and engineering coherence scalars)**.** *The capacity-weighted observer scalar is*

$$\mathbb{C}_U(t;h) = N_U(t)\, \mathfrak{c}_U(t;h). \tag{5.6}$$

The canonical per-channel scalar is

$$\boxed{S_{\rm coh}^{\rm can}(U,t;h) = \frac{\mathbb{C}_U(t;h)}{N_U(t)} = \mathfrak{c}_U(t;h).} \tag{5.7}$$

Under coherent-matter same-channel forcing and Scalar Edge-Center Exhaustion, Section 7 routes $S_{\rm coh}^{\rm can} \in [0,1]$ into the edge-center scalar-slot register. Without that source lift, $S_{\rm coh}^{\rm can}$ is only an operational observer receipt. Let $u_{\rm stored}(U,t)$ be the stored coherent matter energy density in the substrate mode supported by U, let u_0 be a fixed reference energy density, and let $\varepsilon_{\rm sub}(U) \in [0,1]$ be the substrate directness factor. The engineering scalar is

$$S_{\rm coh}^{\rm eng}(U,t;h) = \varepsilon_{\rm sub}(U)\frac{u_{\rm stored}(U,t)}{u_0} S_{\rm coh}^{\rm can}(U,t;h) = N_{\rm coh}(U,t) S_{\rm coh}^{\rm can}(U,t;h), \tag{5.8}$$

where

$$\boxed{N_{\rm coh}(U,t) = \varepsilon_{\rm sub}(U)\frac{u_{\rm stored}(U,t)}{u_0}.} \tag{5.9}$$

Given a partition of unity $w_U(x)$ over observer-supporting subfederations, define the coarse-grained field

$$S_{\rm coh}^{\rm can}(x,t) = \sum_U w_U(x) S_{\rm coh}^{\rm can}(U,t;h), \qquad S_{\rm coh}^{\rm eng}(x,t) = \sum_U w_U(x) S_{\rm coh}^{\rm eng}(U,t;h), \qquad \sum_U w_U(x) = 1. \tag{5.10}$$

With the source lift and linear weak-field response, the response law can be written in either chart:

$$\boxed{\nu_{\rm eff}(x,t) = \nu_{\rm OPH}(x,t) + \chi_\nu^{\rm can} S_{\rm coh}^{\rm can}(x,t) = \nu_{\rm OPH}(x,t) + \chi_\nu^{\rm eng} S_{\rm coh}^{\rm eng}(x,t).} \tag{5.11}$$

On a region with representative $N_{\rm coh}$,

$$\boxed{\chi_\nu^{\rm eng} = \frac{\chi_\nu^{\rm can}}{N_{\rm coh}}.} \tag{5.12}$$

6 Structural Properties

Proposition 6.1 (Observer-facing quotient well-definedness)**.** *Assume two microscopic representatives induce the same observer-accessible continuation law on U, preserve the same visible mismatch score, and expose the same stored coherent energy density and substrate class. Then they give the same $S_{\rm coh}^{\rm can}(U,t;h)$ and $S_{\rm coh}^{\rm eng}(U,t;h)$.*

Proof. The factors R_U, P_U, and C_U are defined from observer-accessible records, boundary-law variables, and visible mismatch data. The channel count N_U cancels in the per-channel scalar. These are precisely the quantities that survive the observer-facing quotient. Multiplying by $u_{\rm stored}$ and $\varepsilon_{\rm sub}$, which are declared observable substrate data for the engineering chart, preserves hidden-representative independence. □

Proposition 6.2 (Null conditions)**.** *If U has no self-read, no stable record algebra, or no predictive boundary coupling at horizon h, then $S_{\rm coh}^{\rm can}(U,t;h) = 0$ and $S_{\rm coh}^{\rm eng}(U,t;h) = 0$.*

Proof. If there is no self-read, then $\mathbf{1}_{\text{self-read}} = 0$. If no stable record algebra is available, then the stability factor R_U vanishes on the declared readout. If predictive boundary coupling is absent, then $P_U = 0$. In each case $\mathfrak{c}_U = 0$, hence both scalars vanish. □

Proposition 6.3 (Exact normalization map)**.** *For the canonical and engineering scalars in Eqs.* (5.7) *and* (5.8)*,*

$$\boxed{\frac{S^{\text{eng}}_{\text{coh}}(U,t;h)}{S^{\text{can}}_{\text{coh}}(U,t;h)} = N_{\text{coh}}(U,t) = \varepsilon_{\text{sub}}(U)\frac{u_{\text{stored}}(U,t)}{u_0}.} \tag{6.1}$$

Proof. Substitute Eq. (5.7) into Eq. (5.8). The quotient is exactly $\varepsilon_{\text{sub}}u_{\text{stored}}/u_0$. □

Proposition 6.4 (Saturated-limit reduction)**.** *If*

$$\mathbf{1}_{\text{self-read}} \simeq 1, \quad R_U \simeq 1, \quad P_U \simeq 1, \quad C_U \simeq 1,$$

then

$$S^{\text{can}}_{\text{coh}}(U,t;h) \simeq 1, \qquad S^{\text{eng}}_{\text{coh}}(U,t;h) \simeq \varepsilon_{\text{sub}}(U)\frac{u_{\text{stored}}(U,t)}{u_0}.$$

Proof. Substitute the saturated factor values into Eqs. (5.7) and (5.8). □

7 Finite Coherent-Matter Source Generator

The channel identity is supplied by importing the finite scalar-slot register, Scalar Edge-Center Exhaustion, and the coherent-matter same-channel theorem from the microphysics paper [4]. The empirical question is separate: whether any real substrate can emit a nonzero, signed, controlled $S^{\text{can}}_{\text{coh}}$ receipt at useful magnitude.

At fixed regulator r, collar or support C, and physical quotient $Q_r = \Sigma_r/\Gamma_r$, write the imported edge-center scalar-slot register as

$$E_{\nu,r}(C), \qquad p_{e,\nu} : Q_r \to \{0,1\}, \qquad \mathcal{N}_{\nu,r,C}(q) = \sum_{e\in E_{\nu,r}(C)} a_e\, p_{e,\nu}(q).$$

Scalar-slot completeness says that, on the quotient-edge scalar branch, the first-order scalar subspace is

$$\mathcal{S}^{(1)}_{\text{sc}}(C) = \operatorname{span}\{\mathcal{N}_{\nu,r,C}\}$$

after constants are removed. Thus a quotient-local, central, edge-center scalar perturbation has no second scalar channel available on that branch.

Definition 7.1 (B.3. Self-reading coherent material subfederation)**.** *Let $U \subset \mathfrak{F}_r$ be an observer-supporting material subfederation with accessible record algebra $\mathcal{Z}^{\text{rec}}_U(t)$, record outcome $Y_U(t)$, future boundary packet $X_{\partial U,t+1:t+h}$, and visible mismatch $\Phi_U(t)$. Its canonical coherent scalar is*

$$S_U(q;t,h) := S^{\text{can}}_{\text{coh}}(U,t;h) = \mathbf{1}_{\text{self-read}}\, R_U(t;h)P_U(t;h)C_U(t).$$

The operational observer-side requirement is the screen-microphysics observer test: bounded accessible interface, durable re-readable records, self-read or internal state estimation, record-conditioned future behavior, boundary prediction better than shuffled controls, and checkpoint continuation within declared error.

Definition 7.2 (B.4. Scalar-slot footprint)**.** *The material subfederation U induces a quotient-visible footprint*

$$b_{U,r,C} : E_{\nu,r}(C) \times Q_r \to \mathbb{R}_{\geq 0}$$

with

$$\sum_{e \in E_{\nu,r}(C)} b_{U,r,C}(e;q) = 1$$

on the active support of U, and $b_{U,r,C}(e;q) = 0$ outside the scalar slots exposed by the visible boundary of U. Define the local material slot count and availability factor

$$\mathcal{N}_{\nu,U}(q) = \sum_{e \in E_{\nu,r}(C)} b_{U,r,C}(e;q) a_e \, p_{e,\nu}(q), \qquad \mathcal{O}_U(q) = \sum_{e \in E_{\nu,r}(C)} b_{U,r,C}(e;q) p_{e,\nu}(q),$$

and

$$A_{U,r,C}(q) = \sum_{e \in E_{\nu,r}(C)} b_{U,r,C}(e;q) a_e \, [1 - p_{e,\nu}(q)].$$

The source is non-saturated at q when $A_{U,r,C}(q) > 0$.

The footprint is not a new field. It is a quotient-visible map from the material boundary to the scalar slots it can touch. If the material has no self-read, no stable record algebra, or no predictive boundary coupling, Proposition 6.2 forces $S^{\mathrm{can}}_{\mathrm{coh}} = 0$.

Definition 7.3 (B.5. Unit-normalized coherent scalar source generator)**.** *For $q \in Q_r$ and $e \in E_{\nu,r}(C)$, define*

$$q \oplus e := n_r(q \text{ with scalar slot } e \text{ activated}),$$

where n_r is the quotient normal-form map. If the slot is active in q, set $q \oplus e = q$. For any test function $f : Q_r \to \mathbb{R}$, define

$$(\mathcal{L}^{\mathrm{coh}}_{U,r,C} f)(q) = S_U(q;t,h) \sum_{e \in E_{\nu,r}(C)} b_{U,r,C}(e;q) \big[f(q \oplus e) - f(q) \big].$$

This is the unit-normalized canonical generator. Engineering inefficiencies, stored-energy amplification, substrate directness, and geometry factors belong to the engineering chart, not to the canonical channel identity.

Lemma 7.4 (B.6. Quotient descent)**.** *$\mathcal{L}^{\mathrm{coh}}_{U,r,C}$ is a well-defined finite generator on the physical quotient Q_r.*

Proof. S_U is built from the self-read gate, record stability, predictive boundary coupling, and visible mismatch data. The footprint $b_{U,r,C}$ is quotient-visible by definition. The update $q \oplus e$ is normal-formed by n_r. Therefore replacing a microscopic representative of q by a gauge-equivalent representative changes none of S_U, $b_{U,r,C}$, $q \oplus e$, or the value of the generator. Since Q_r and $E_{\nu,r}(C)$ are finite, the generator is finite. □

Theorem 7.5 (B.7. Coherent material perturbs the canonical scalar count)**.** *Assume the finite scalar-slot register, scalar-slot completeness, a coherent material footprint, and the source generator above. If $S_U(q;t,h) > 0$ and $A_{U,r,C}(q) > 0$, then the coherent material subfederation U perturbs the same quotient-edge scalar opportunity count $\mathcal{N}_{\nu,r,C}$ that appears in the dark-sector collar channel:*

$$(\mathcal{L}^{\mathrm{coh}}_{U,r,C} \mathcal{N}_{\nu,r,C})(q) = S_U(q;t,h) A_{U,r,C}(q) > 0.$$

By scalar-slot completeness, the scalar projection of the coherent-matter perturbation lies in

$$\mathrm{span}\{\mathcal{N}_{\nu,r,C}\},$$

so it cannot define an independent scalar channel.

Proof. Evaluate the generator on $\mathcal{N}_{\nu,r,C}$:

$$(\mathcal{L}^{\mathrm{coh}}_{U,r,C}\mathcal{N}_{\nu,r,C})(q) = S_U(q;t,h)\sum_e b_{U,r,C}(e;q)\left[\mathcal{N}_{\nu,r,C}(q\oplus e) - \mathcal{N}_{\nu,r,C}(q)\right].$$

By definition of $q \oplus e$, activating an inactive slot e increases $\mathcal{N}_{\nu,r,C}$ by a_e. A slot active in q contributes zero:

$$\mathcal{N}_{\nu,r,C}(q\oplus e) - \mathcal{N}_{\nu,r,C}(q) = a_e[1 - p_{e,\nu}(q)].$$

Substitution gives

$$(\mathcal{L}^{\mathrm{coh}}_{U,r,C}\mathcal{N}_{\nu,r,C})(q) = S_U(q;t,h)\sum_e b_{U,r,C}(e;q)a_e[1 - p_{e,\nu}(q)] = S_U(q;t,h)A_{U,r,C}(q).$$

The positivity follows from the two nonzero hypotheses. The generator changes only the central edge-center scalar-slot indicators $p_{e,\nu}$. By scalar-slot completeness, the first-order scalar projection is generated by $\mathcal{N}_{\nu,r,C}$, with constants removed. Hence the material perturbation acts in the canonical quotient-edge scalar channel itself. □

Definition 7.6 (B.8. Oriented coherent scalar contrast)**.** *For a two-zone device or material body, let U_+ and U_- be the two observer-supporting subfederations separated by the declared vertical orientation. In the lab force convention used below, U_+ is the bottom zone and U_- is the top zone. Define*

$$\Delta S^{\mathrm{can}}_{\mathrm{coh}} = S^{\mathrm{can}}_{\mathrm{coh}}(U_+,t;h) - S^{\mathrm{can}}_{\mathrm{coh}}(U_-,t;h).$$

The signed source generator is

$$\mathcal{L}^{\mathrm{coh}}_{\Delta} = \mathcal{L}^{\mathrm{coh}}_{U_+} - \mathcal{L}^{\mathrm{coh}}_{U_-},$$

with scalar projection

$$\mathcal{L}^{\mathrm{coh}}_{\Delta}\mathcal{N}_\nu = S_{U_+}A_{U_+} - S_{U_-}A_{U_-}.$$

On a symmetric footprint branch where $A_{U_+} = A_{U_-} = 1$, this reduces to

$$\mathcal{L}^{\mathrm{coh}}_{\Delta}\mathcal{N}_\nu = \Delta S^{\mathrm{can}}_{\mathrm{coh}}.$$

Definition 7.7 (B.9. Coherent scalar source action)**.** *Let $\mu_{0,r}$ be the unperturbed quotient law on Q_r, with action $I_{0,r}$. Define*

$$I_{\eta,r,U}(q) = I_{0,r}(q) - \eta\, S_U(q;t,h)\mathcal{N}_{\nu,U}(q).$$

The source-perturbed finite quotient law is

$$\mu_{\eta,r,U}(q) = Z_\eta^{-1}\exp[-I_{\eta,r,U}(q)].$$

Proposition 7.8 (B.10. Linear response through the canonical scalar count)**.** *For any bounded quotient observable $F : Q_r \to \mathbb{R}$,*

$$\frac{\mathrm{d}}{\mathrm{d}\eta}\mathbb{E}_\eta[F]\Big|_{\eta=0} = \mathrm{Cov}_0\left(F,\, S_U \mathcal{N}_{\nu,U}\right).$$

In particular,

$$\frac{\mathrm{d}}{\mathrm{d}\eta}\mathbb{E}_\eta[\mathcal{N}_{\nu,r,C}]\Big|_{\eta=0} = \mathrm{Cov}_0\left(\mathcal{N}_{\nu,r,C},\, S_U \mathcal{N}_{\nu,U}\right).$$

On the nondegenerate scalar branch this covariance is nonzero whenever S_U is nonzero, the footprint is nonzero, and $\mathcal{N}_{\nu,U}$ is not μ_0-almost surely constant.

Proof. Differentiate the finite partition function

$$\mathbb{E}_\eta[F] = \frac{\sum_q F(q) e^{-I_{0,r}(q)+\eta S_U(q;t,h)\mathcal{N}_{\nu,U}(q)}}{\sum_q e^{-I_{0,r}(q)+\eta S_U(q;t,h)\mathcal{N}_{\nu,U}(q)}}.$$

At $\eta = 0$, the numerator derivative gives $\mathbb{E}_0[F\, S_U \mathcal{N}_{\nu,U}]$, and the partition-function derivative subtracts $\mathbb{E}_0[F]\mathbb{E}_0[S_U \mathcal{N}_{\nu,U}]$. This is the displayed covariance. Since $\mathcal{N}_{\nu,U}$ is built from the same $p_{e,\nu}$ slots, scalar-slot completeness again puts the scalar response in the $\mathcal{N}_{\nu,r,C}$ channel. □

Corollary 7.9 (B.11. Tier-B response law from the finite generator)**.** *On the finite quotient-edge branch above, coherent matter changes the canonical scalar opportunity count by a term proportional to $S^{\mathrm{can}}_{\mathrm{coh}}$. Linear weak-field response therefore has the Tier-B1 form*

$$\delta\nu = \chi^{\mathrm{can}}_\nu S^{\mathrm{can}}_{\mathrm{coh}}.$$

If the Tier-C co-registered collar survival assumptions are also granted, then

$$\chi^{\mathrm{can}}_\nu = \lambda_{\mathrm{collar}}, \qquad \nu_{\mathrm{eff}} = \nu_{\mathrm{OPH}} + \lambda_{\mathrm{collar}} S^{\mathrm{can}}_{\mathrm{coh}}.$$

In the stored-energy engineering chart,

$$S^{\mathrm{eng}}_{\mathrm{coh}} = N_{\mathrm{coh}} S^{\mathrm{can}}_{\mathrm{coh}}, \qquad \chi^{\mathrm{eng}}_\nu = \frac{\chi^{\mathrm{can}}_\nu}{N_{\mathrm{coh}}}, \qquad \nu_{\mathrm{eff}} = \nu_{\mathrm{OPH}} + \chi^{\mathrm{eng}}_\nu S^{\mathrm{eng}}_{\mathrm{coh}}.$$

Proof. Theorem 7.5 proves that the material source perturbs the same quotient-edge scalar count used by the dark-sector collar channel. Linear weak-field response makes the induced perturbation of ν proportional to that count in canonical normalization. The Tier-C collar theorem identifies the proportionality coefficient with the collar survival coefficient. The engineering formulas are the chart change $S^{\mathrm{eng}}_{\mathrm{coh}} = N_{\mathrm{coh}} S^{\mathrm{can}}_{\mathrm{coh}}$. □

Proposition 7.10 (Finite channel bridge: slots and slices are one family)**.** *Let E be one finite indexed family equipped with both a record panel*

$$(p_e(q), a_e, q \mapsto q \oplus e)_{e\in E}$$

and a collar panel

$$(w_e, \epsilon_e)_{e\in E}, \qquad w_e \geq 0, \quad \sum_{e\in E} w_e = 1, \quad \epsilon_e \geq 0.$$

The record panel defines the scalar-slot register of Definition 7.2 and the coherent generator of Definition 7.3. The collar panel defines the finite-slice survival coefficient

$$\lambda_{\mathrm{collar}} = \sum_{e\in E} w_e \, e^{-\epsilon_e}.$$

Because both panels use the same index family E, the statement that coherent matter perturbs the same counter priced by the collar is not an extra equality between two unrelated constructions. It is a single-channel structure.

Corollary 7.11 (Composite Tier-B1 channel law)**.** *Under the hypotheses of Theorem 7.5, and under the uniform product-thickening clause $\epsilon_e = P/24$ for every $e \in E$, the record-source increment and the collar survival factor compose as*

$$\lambda_{\mathrm{collar}} \, (\mathcal{L}^{\mathrm{coh}}_{U,r,C}\mathcal{N}_{\nu,r,C})(q) = e^{-P/24} \, S_U(q;t,h) \, A_{U,r,C}(q).$$

Proof. Theorem 7.5 gives

$$(\mathcal{L}^{\mathrm{coh}}_{U,r,C}\mathcal{N}_{\nu,r,C})(q) = S_U(q;t,h) \, A_{U,r,C}(q).$$

The finite collar gate gives $\lambda_{\mathrm{collar}} = e^{-P/24}$ under slice-wise reserve unbiasedness. Multiplying the two identities gives the displayed composite law. □

Remark 7.12 (What the channel bridge does not close)*.* Proposition 7.10 removes a bookkeeping ambiguity: the record slots and collar slices can be represented as one finite channel. Two physical residues remain. First, an actual material system must supply receipts showing that its record channel and collar channel instantiate this one structure rather than two unrelated counters. Second, the numerical record-ΔS to gravity-ΔS calibration, denoted G9 in the audit ledger, remains an external calibration. A null experiment therefore bounds the product $\chi_\nu \Delta S$, not χ_ν and ΔS separately.

Remark 7.13 (Generator realization is not a law selector)*.* The generator above is a realization after the quotient branch, scalar-slot register, and source law have been selected. It is not a selector of a unique physical probability law by itself. The probability-law statement is the finite action formulation in Definition 7.7.

7.1 Source-generator proof ledger

Target	Closure	Status
Coherent matter sources same scalar channel	Finite source generator $\mathcal{L}^{\mathrm{coh}}_{U,r,C}$ acts on edge-center scalar slots; scalar-slot completeness makes the scalar projection rank-one, generated by $\mathcal{N}_{\nu,r,C}$.	Theorem-level on the finite quotient-edge branch; empirical nonzero $S^{\mathrm{can}}_{\mathrm{coh}}$ is a hardware receipt.
Finite channel bridge	The record-side source generator and collar-side reserve coefficient are represented by one finite indexed family of slots/slices, yielding the composite law of Corollary 7.11.	Bookkeeping theorem; nature-instantiation and G9 numerical calibration remain physical obligations.
Switchable-force energy cage	Any conservative ABBA cycle obeys $W_{\mathrm{ABBA}} = \mathrm{Toggle}(q_1) - \mathrm{Toggle}(q_2)$.	Theorem-level accounting constraint; the local response law Eq. (9.3) remains a branch law, not derived local dynamics.

7.2 Deterministic source-generator receipts

Implementation evidence for this branch should expose the following deterministic receipt names.

`SELF_READ_PATCH_RECEIPT`
Passes when U satisfies the operational observer test: bounded interface, durable re-readable records, self-read or internal state estimation, record-conditioned future behavior, boundary prediction better than shuffled controls, and checkpoint continuation.

`SCOH_CANONICAL_RECEIPT`
Emits $R_U(t;h)$, $P_U(t;h)$, $C_U(t)$, $\mathbf{1}_{\text{self-read}}$, $N_U(t)$, and $S^{\mathrm{can}}_{\mathrm{coh}}$, with record-algebra hash, boundary-packet hash, mismatch-functional hash, and shuffled-record controls.

`EDGE_SOURCE_LIFT_RECEIPT`
Emits the footprint $b_{U,r,C}$, proves it factors through the quotient-edge scalar register, and proves invariance under port relabeling, hidden carrier coordinates, mesh labels, and accepted repair schedule changes.

`FINITE_CHANNEL_BRIDGE_RECEIPT`
Emits the shared finite index family E and its two panels: record activity, opportunity weights, and activation maps on the source side; slice weights and protected-reserve means on the collar side. Fails if the source generator and collar survival factor are computed over unrelated counters.

`NO_SEPARATE_SCALAR_CHANNEL_RECEIPT`
Fails if the coherent material source depends on bulk labels, screen-area labels, noncentral collar variables, heat, rest mass, electrical power, or any presentation variable not carried by the scalar-slot register.

`SOURCE_NONDEGENERACY_RECEIPT`
: Checks

$$\mathrm{Cov}_0(\mathcal{N}_{\nu,r,C}, S_U \mathcal{N}_{\nu,U}) \neq 0$$

or reports the selected ensemble as source-degenerate.

`VERTICAL_CONTRAST_RECEIPT`
: Emits the signed bottom-minus-top $\Delta S^{\mathrm{can}}_{\mathrm{coh}}$, plus sign-reversal runs, sham runs, thermal/electromagnetic/mechanical controls, and record-shuffle controls.

`FINITE_GENERATOR_REALIZATION_RECEIPT`
: Optional. Emits the slot update maps or repair projections implementing $\mathcal{L}^{\mathrm{coh}}_{U,r,C}$, while labeling the generator as a realization of an externally selected source law.

`DARK_SECTOR_SIMULATION_PLAN_RECEIPT`
: Optional integration receipt. It records which dark-sector simulator gates are present across the static galaxy, finite covariant parent, finite-collar Boltzmann, Boltzmann-input, anomaly, and frozen-likelihood reports, and names the next blocked command. It is not a force receipt, not a susceptibility measurement, and not a replacement for the source-lift receipts above.

8 Canonical Quotient-Edge Susceptibility

This section is conditional on the closed quotient-edge collar branch. The recovered OPH core and the coherent-matter same-channel theorem supply the observer-facing carrier, the coherent scalar source readout, and the scalar channel. The nonzero coefficient follows after the collar lemmas below are granted.

Assumption 8.1 (Imported quotient-edge collar theorem stack)**.** *The χ_ν branch imports the following theorem-level source package from the screen microphysics paper [4]:*

`def:oph-coherent-material-subfederation`, `def:canonical-coherent-matter-scalar-source`, `lem:coherent-material-source-quotient-descent`, `thm:nonzero-coherent-scalar-source`, `thm:coherent-matter-same-channel-forcing`, `thm:unique-scalar-linear-response`.

These imports state that a nondegenerate OPH-coherent material subfederation has

$$S^{\mathrm{can}}_{\mathrm{coh}} = \mathbf{1}_{\mathrm{self-read}}\, \mathsf{R}_U\, \mathsf{P}_U\, \mathsf{C}_U > 0$$

and that this source lies in the same edge-center scalar register as the protected Z_6 collar channel under Scalar Edge-Center Exhaustion. The collar coefficient uses the following branch-local inputs.

L1. Positive scalar source branch: the scalar repair defect is the minimal recoverability defect

$$R_C = I(A : D \mid B) \geq 0.$$

L2. Quotient-edge scalar locality: every quotient-local scalar recoverability event is resolved on the edge-center cut register. This is imported from the microphysics theorem labeled `thm:scalar-activation-edge-center`, under the branch assumption labeled `ass:edge-center-scalar-completeness` [4].

L3. Co-registration with the protected reserve: scalar activation and the protected $\mathbb{Z}_6$ reserve live on the same edge-center resolution and commute. This is imported from the microphysics theorem labeled `thm:scalar-z6-same-register-commute`.

L4. Scalar-channel exhaustion and no separate scalar carrier: the scalar activation channel is exhausted by the scalar activation projector, and any separate bulk, screen-area, noncentral, or hidden-carrier scalar channel either changes quotient observables or is inert. This is imported from the microphysics theorems labeled `thm:scalar-channel-exhaustion` and `thm:no-separate-scalar-carrier`.

L5. Local Poisson reserve survival: on a co-registered scalar slot at transverse coordinate y,

$$\lambda_{\mathrm{slot}}(y) = \exp[-\epsilon_{\mathbb{Z}_6}(y)].$$

This is a declared Poisson branch unless separately derived from a finite MaxEnt/occupancy theorem.

L6. Finite-thickness averaging with scalar-weighted reserve mean:

$$\lambda_{\mathrm{collar}} = \int \mathrm{d}y\, w(y) \exp[-\epsilon_{\mathbb{Z}_6}(y)],$$

with

$$\int \mathrm{d}y\, w(y)\epsilon_{\mathbb{Z}_6}(y) = \frac{P}{24}.$$

The profile definitions, mean-reserve disintegration, and survival formula are imported from the microphysics definitions and theorems labeled `def:finite-thickness-scalar-profile`, `thm:finite-thickness-mean-reserve`, and `thm:finite-thickness-collar-survival`.

L7. Exact uniform branch: on the uniform product-thickening branch, including product algebra, product trace, reserve pullback, scalar activation disintegration, and slice-wise scalar-reserve unbiasedness,

$$\lambda_{\mathrm{collar}} = e^{-P/24}.$$

This is imported from the microphysics definition and theorem labeled `def:uniform-product-thickening-bra` and `thm:exact-uniform-product-thickening-coefficient`.

The public OPH readout gives

$$P = 1.630968209403959. \tag{8.1}$$

The normalized quotient reserve mean is

$$\boxed{\tau_q(Z_{6,r,C}) = \frac{P}{24} = 0.06795700872516496.} \tag{8.2}$$

The reciprocal reserve-depth trace is

$$\boxed{\mathrm{Tr}_q^{\#}(Z_{6,r,C}) = \frac{24}{P} = 14.715185655746689.} \tag{8.3}$$

Only the normalized mean $P/24$ enters the Poisson exponent.

Theorem 8.2 (Forced canonical susceptibility on the co-registered branch)**.** *Assume L1 to L6 of Assumption 8.1. Then the canonically normalized coherent-matter susceptibility is the collar survival coefficient:*

$$\boxed{\chi_\nu^{\mathrm{can}} = \lambda_{\mathrm{collar}}.} \tag{8.4}$$

Proof. By Theorem 7.5 and Corollary 7.9, $S_{\mathrm{coh}}^{\mathrm{can}}$ enters the same quotient-edge scalar opportunity count before the protected reserve acts. By L3, the $\mathbb{Z}_6$ reserve acts on that same register. By L5, each local coherent scalar opportunity survives the reserve with factor $\exp[-\epsilon_{\mathbb{Z}_6}(y)]$. By L6, finite-thickness averaging gives the surviving coherent scalar fraction

$$S_{\mathrm{coh,after}}^{\mathrm{can}} = \lambda_{\mathrm{collar}} S_{\mathrm{coh,before}}^{\mathrm{can}}.$$

The continuation law writes the same surviving fraction as

$$S_{\mathrm{coh,after}}^{\mathrm{can}} = \chi_\nu^{\mathrm{can}} S_{\mathrm{coh,before}}^{\mathrm{can}}.$$

For any nonzero canonical coherent fraction, the coefficients are equal. □

Corollary 8.3 (Theorem-level nonzero band)**.** *Under L1 to L6,*

$$\boxed{0.9343006394893864\ldots \leq \chi_\nu^{\mathrm{can}} \leq 1.} \tag{8.5}$$

Proof. The function e^{-x} is convex, so the Jensen inequality gives

$$\chi_\nu^{\mathrm{can}} = \int \mathrm{d}y\, w(y) e^{-\epsilon_{\mathbb{Z}_6}(y)} \geq \exp\Big[-\int \mathrm{d}y\, w(y)\epsilon_{\mathbb{Z}_6}(y)\Big] = e^{-P/24}.$$

Using Eq. (8.2),

$$e^{-P/24} = 0.9343006394893864\ldots.$$

Reserve survival factors are at most one, so $\chi_\nu^{\mathrm{can}} \leq 1$. □

Corollary 8.4 (Exact canonical value on the uniform branch)**.** *Assume L7 in addition to L1 to L6. Then*

$$\boxed{\chi_\nu^{\mathrm{can}} = e^{-P/24} = 0.9343006394893864\ldots.} \tag{8.6}$$

Proof. By L7, $\lambda_{\mathrm{collar}} = e^{-P/24}$. Use Eq. (8.4). □

Remark 8.5 (Uniformity is stronger than product trace). The exact value $e^{-P/24}$ does not follow from the unconditioned trace $\tau_q(Z_6) = P/24$ alone. It requires the scalar-active edge-slot sampler to be reserve-unbiased. Equivalently, for every scalar-active transverse slice,

$$\frac{\tau(\Pi^{\mathrm{scal}}(y) Z_6)}{\tau(\Pi^{\mathrm{scal}}(y))} = \frac{P}{24}.$$

If scalar activation is correlated with the protected reserve inside the edge-center algebra, then $\epsilon_{\mathbb{Z}_6}(y)$ may vary and the exact coefficient is not licensed. The valid coefficient is then the profile integral

$$\lambda_{\mathrm{collar}} = \int \mathrm{d}y\, w(y) e^{-\epsilon_{\mathbb{Z}_6}(y)}.$$

Remark 8.6 (Profile-envelope number). The dark-sector correction note also lists profile targets: best same-function RAR 0.958802256, binned RAR 0.936737944, and the OPH $\mathbb{Z}_6$/Poisson value 0.934300639. If an extra numerical selector keeps the coherent continuation inside that same-function profile envelope, one may quote the practical profile range

$$0.934300639 \le \chi_\nu^{\mathrm{can}} \le 0.958802256.$$

The range is selector-dependent. The theorem-level band used in the rest of this paper is Eq. (8.5), with the exact value Eq. (8.6) on L7.

Corollary 8.7 (Engineering chart band)**.** *On any engineering chart with representative normalization* N_{coh}*,*

$$\boxed{\frac{0.93430063948938 64}{N_{\mathrm{coh}}} \le \chi_\nu^{\mathrm{eng}} \le \frac{1}{N_{\mathrm{coh}}}.} \tag{8.7}$$

On the exact uniform branch,

$$\boxed{\chi_\nu^{\mathrm{eng}} = \frac{e^{-P/24}}{N_{\mathrm{coh}}}.} \tag{8.8}$$

N_{coh}	theorem-level χ_ν^{eng} band	exact-branch χ_ν^{eng}
10^{20}	9.34×10^{-21} to 1.00×10^{-20}	9.34×10^{-21}
10^{21}	9.34×10^{-22} to 1.00×10^{-21}	9.34×10^{-22}
10^{22}	9.34×10^{-23} to 1.00×10^{-22}	9.34×10^{-23}
10^{23}	9.34×10^{-24} to 1.00×10^{-23}	9.34×10^{-24}
10^{24}	9.34×10^{-25} to 1.00×10^{-24}	9.34×10^{-25}

Table 2: Translation from the theorem-level canonical band to the engineering chart. Here $N_{\mathrm{coh}} = \varepsilon_{\mathrm{sub}} u_{\mathrm{stored}}/u_0$.

9 Weak-Field Response

Insert Eq. (5.11) into the settled weak-field density:

$$\rho_A^{\mathrm{eff}} = -\frac{1}{4\pi G}\nabla \cdot \Big[\big(\nu_{\mathrm{OPH}} - 1 + \chi_\nu^{\mathrm{eng}} S_{\mathrm{coh}}^{\mathrm{eng}}\big)\mathbf{g}_b\Big]. \tag{9.1}$$

In a small terrestrial laboratory region with $\mathbf{g}_b \simeq -g\,\hat{z}$ and canonical $\nu_{\mathrm{OPH}} - 1$ negligible, this becomes

$$\rho_A^{\mathrm{lab}} \simeq -\frac{1}{4\pi G}\nabla \cdot \big[\chi_\nu^{\mathrm{eng}} S_{\mathrm{coh}}^{\mathrm{eng}}\,\mathbf{g}\big]. \tag{9.2}$$

Equation (9.2) uses S_{coh} only after the coherent material source-generator theorem has routed S_{coh} into the quotient-edge scalar opportunity count. Heat, rest mass, and ordinary stored energy do not enter unless they pass the self-read, record, prediction, and quotient-edge footprint receipts.

Theorem 9.1 (Planar response law)**.** *Let a thin device have horizontal projected area* $A_\perp$*, and define the bottom-to-top coherence contrast*

$$\Delta S_{\mathrm{coh}}^{\mathrm{eng}} = S_{\mathrm{coh\,bottom}}^{\mathrm{eng}} - S_{\mathrm{coh\,top}}^{\mathrm{eng}}.$$

Then, to leading order in the thin-device approximation, the apparent vertical force generated by the engineering-chart continuation is

$$\boxed{F_\chi = \frac{g^2}{4\pi G} A_\perp \, \chi_\nu^{\text{eng}} \, \Delta S_{\text{coh}}^{\text{eng}}.} \tag{9.3}$$

Proof. With $\mathbf{g} = -g\hat{z}$, Eq. (9.2) gives

$$\rho_A^{\text{lab}} = \frac{g\chi_\nu^{\text{eng}}}{4\pi G} \frac{\partial S_{\text{coh}}^{\text{eng}}}{\partial z}$$

up to the sign convention for z. Integrating through a thin column gives

$$M_A = \int \rho_A^{\text{lab}} \, \mathrm{d}V = -\frac{g\chi_\nu^{\text{eng}}}{4\pi G} A_\perp \, \Delta S_{\text{coh}}^{\text{eng}}$$

when $\Delta S_{\text{coh}}^{\text{eng}}$ is bottom minus top. The apparent weight change is $F_\chi = -gM_A$, yielding Eq. (9.3). □

Corollary 9.2 (Geometric ceiling)**.** *If $|\Delta\nu| \lesssim 1$, then*

$$\frac{|F_\chi|}{A_\perp} \lesssim C_g \equiv \frac{g^2}{4\pi G} = 1.1466 \times 10^{11} \, \mathrm{N \, m^{-2}} \tag{9.4}$$

for $g = 9.80665 \, \mathrm{m \, s^{-2}}$ and $G = 6.67430 \times 10^{-11} \, \mathrm{m^3 \, kg^{-1} \, s^{-2}}$.

The ceiling is algebraic. Material reachability of $|\Delta\nu| = 1$ is a separate claim.

10 Engineering Susceptibility Window

For bounds it is useful to package the contrast geometry, substrate directness, and OPH coherence factors into one dimensionless number:

$$\Delta S_{\text{coh}}^{\text{eng}} = \Gamma_{\text{eff}} \frac{u}{u_0}, \qquad 0 \le \Gamma_{\text{eff}} \le 1. \tag{10.1}$$

Here u is the representative stored coherent energy density in the active mode. The factor Γ_{eff} includes vertical contrast, substrate directness, and effective canonical observer coherence.

Theorem 10.1 (Nonempty macroscopic linear-response window)**.** *Let $\Sigma = M/A_\perp$ be the areal load, let $f \in [0,1]$ be the target fraction of the ordinary weight Mg, and let $\epsilon_{\text{lin}} \le 1$ be the maximum allowed $|\Delta\nu|$ in the chosen linear-response regime. Assume $\Gamma_{\text{eff}} > 0$. If*

$$f \, \frac{4\pi G \Sigma}{g} < \epsilon_{\text{lin}}, \tag{10.2}$$

then the interval

$$\boxed{\frac{4\pi G \Sigma f}{g} \frac{u_0}{\Gamma_{\text{eff}} u} \le |\chi_\nu^{\text{eng}}| \le \epsilon_{\text{lin}} \frac{u_0}{\Gamma_{\text{eff}} u}} \tag{10.3}$$

is nonempty. Values in this interval produce at least fraction f of the ordinary weight inside the declared linear-response bound. Conversely, any value that produces at least fraction f while staying inside that linear-response bound lies in this interval.

Proof. The force law gives

$$\frac{|F_\chi|}{Mg} = \frac{g}{4\pi G\Sigma}|\chi_\nu^{\rm eng}|\Delta S_{\rm coh}^{\rm eng}.$$

Requiring $|F_\chi|/(Mg) \geq f$ gives the lower bound in Eq. (10.3). Requiring $|\Delta\nu| = |\chi_\nu^{\rm eng}|\Delta S_{\rm coh}^{\rm eng} \leq \epsilon_{\rm lin}$ gives the upper bound. The interval is nonempty exactly when the lower bound is smaller than the upper bound, which is Eq. (10.2). □

Corollary 10.2 (Numerical width in normalized areal load)**.** *Let*

$$\Sigma_2 = \frac{\Sigma}{10^2\,\mathrm{kg\,m^{-2}}}.$$

Then

$$\frac{4\pi G\Sigma}{g} = 8.5525 \times 10^{-9}\,\Sigma_2.$$

The full-response ($f = 1$)*, order-unity linear window has width*

$$\frac{\chi_{\rm upper}}{\chi_{\rm lower}} = 1.1692 \times 10^8\,\Sigma_2^{-1},$$

independent of u *and* $\Gamma_{\rm eff}$*. With* $\epsilon_{\rm lin} = 0.1$*, the window width is* $1.1692 \times 10^7\,\Sigma_2^{-1}$.

$u\,[\mathrm{J\,m^{-3}}]$	u/u_0	$\lvert\chi_\nu^{\rm eng}\rvert_{\rm req}$ for $f\Sigma_2/\Gamma_{\rm eff} = 1$	$\lvert\chi_\nu^{\rm eng}\rvert_{\rm lin}$ for $\epsilon_{\rm lin}/\Gamma_{\rm eff} = 1$
1.0×10^5	1.18×10^{14}	7.27×10^{-23}	8.5×10^{-15}
5.5×10^5	6.47×10^{14}	1.32×10^{-23}	1.55×10^{-15}
1.0×10^6	1.18×10^{15}	7.27×10^{-24}	8.5×10^{-16}
4.0×10^6	4.71×10^{15}	1.82×10^{-24}	2.13×10^{-16}
1.0×10^7	1.18×10^{16}	7.27×10^{-25}	8.5×10^{-17}

Table 3: Full-response susceptibility bounds for $u_0 = 8.5 \times 10^{-10}\,\mathrm{J\,m^{-3}}$ and $\Sigma_2 = \Sigma/(10^2\,\mathrm{kg\,m^{-2}})$. The third column should be multiplied by $f\Sigma_2/\Gamma_{\rm eff}$. The fourth column should be multiplied by $\epsilon_{\rm lin}/\Gamma_{\rm eff}$.

The table gives the bound in numerical form. It says that ordinary material stress and coherent phonon energy densities put the macroscopic response threshold in the $\chi_\nu^{\rm eng} \sim 10^{-22}$ to 10^{-24} range for macroscopic areal loads and $\Gamma_{\rm eff}$ near unity. Smaller $\Gamma_{\rm eff}$ shifts the required value upward by $1/\Gamma_{\rm eff}$.

11 Power and Stored-Energy Bounds

11.1 Conservation-law cage for switchable force claims

The force law fixes stored energy. Input power depends on how the coherent state is maintained. A stronger bookkeeping constraint applies to any switchable force claim. Let q denote a device position and $s \in \{\text{off}, \text{on}\}$ its internal state. In any conservative sector with state energy $E(q, s)$, define the toggle ledger

$$\text{Toggle}(q) := E(q, \text{on}) - E(q, \text{off})$$

and the work extracted by the ABBA cycle

$$(\text{on at } q_1) \to (q_2, \text{on}) \to (\text{off at } q_2) \to (q_1, \text{off}).$$

Then

$$W_{\text{ABBA}} = \text{Toggle}(q_1) - \text{Toggle}(q_2).$$

Thus a claimed position-dependent switchable force with no toggle-energy ledger is a perpetual-motion claim: if the toggle cost is zero everywhere, the cycle work is zero. If the cycle extracts $W > 0$, some position pair must carry a toggle-ledger difference of at least W, and at least one endpoint must carry a ledger entry of order $W/2$ or larger relative to the pair.

This is the G10 conservation-law cage. It prices any positive χ_ν detection, but it does not derive the response law Eq. (9.3). A detection must therefore arrive with both the ordinary force controls and a toggle/maintenance energy ledger large enough to account for the measured cycle work.

11.2 Driven coherent oscillations

For a driven oscillator with angular frequency ω, mode volume V_{mode}, quality factor Q, and stored density u,

$$P_{\text{loss}} = \frac{u V_{\text{mode}} \omega}{Q}. \tag{11.1}$$

Combining Eq. (11.1) with the required density from the macroscopic response law gives

$$\boxed{P_{\text{loss}} = \frac{4\pi G \Sigma f\, u_0\, V_{\text{mode}} \omega}{g\, Q\, \Gamma_{\text{eff}}\, |\chi_\nu^{\text{eng}}|}}. \tag{11.2}$$

For $\Sigma_2 = 1$, $V_{\text{mode}} = 10^{-3}\,\text{m}^3$, $\omega = 2\pi(4.0 \times 10^4)\,\text{s}^{-1}$, $Q = 10^5$, $f = 1$, and $\Gamma_{\text{eff}} = 1$, this gives:

$\lvert\chi_\nu^{\text{eng}}\rvert$	P_{loss}
10^{-20}	1.8 W
10^{-21}	18 W
10^{-22}	183 W
10^{-23}	1.8 kW
10^{-24}	18 kW

The scaling is $P_{\text{loss}} \propto (Q\Gamma_{\text{eff}}|\chi_\nu^{\text{eng}}|)^{-1}$.

11.3 Static elastic storage

For a static elastic-strain coherent variable, the stored density is

$$u_{\text{elastic}} = \frac{\sigma^2}{2E}, \tag{11.3}$$

where σ is stress and E is the Young modulus. In that interpretation there is no oscillator decay factor ω/Q. Maintaining the state is then limited by creep, relaxation, and mechanical failure. Its loss law differs from Eq. (11.1). The same susceptibility bounds apply, because they depend only on u, Γ_{eff}, and Σ.

12 Construction-Scale Consequences

The bounds are algebraic. Substrate reachability of $\Gamma_{\text{eff}} \simeq 1$ and safe maintenance of a high-energy coherent state are separate engineering claims. The bounds specify the requirements for a macroscopic force device if the continuation is real.

A force device, in this narrow sense, is a bounded substrate whose active region creates a vertical engineering-coherence contrast $\Delta S_{\text{coh}}^{\text{eng}}$. The apparent weight fraction is

$$\frac{|F_\chi|}{Mg} = \frac{g}{4\pi G\Sigma}\,|\chi_\nu^{\text{eng}}|\,\Gamma_{\text{eff}}\,\frac{u}{u_0}. \tag{12.1}$$

All construction choices enter through the product

$$|\chi_\nu^{\text{eng}}|\,\Gamma_{\text{eff}}\,u. \tag{12.2}$$

The product makes χ_ν^{eng} the governing chart number. Geometry fixes the coefficient $g/(4\pi G\Sigma)$. Better materials can raise u. Better mode design can raise Γ_{eff}. A smaller χ_ν^{eng} must be paid for linearly by higher stored energy density, larger active volume, better quality factor, or a smaller target response fraction. If $\chi_\nu^{\text{eng}} = 0$, no amount of construction produces the continuation force.

For the normalized areal load $\Sigma_2 = 1$, the full-response condition is

$$\Delta\nu_{\text{full}}(\Sigma_2 = 1) = \frac{4\pi G\Sigma}{g} = 8.5525 \times 10^{-9}. \tag{12.3}$$

The value is far below even a conservative $\epsilon_{\text{lin}} = 0.1$ linear bound. The required $\Delta\nu$ lies inside the theorem window. Construction depends on the required $\Gamma_{\text{eff}}u$ product, mode volume, quality factor, and material stress.

Two consequences follow.

First, a measurement of χ_ν^{eng} selects a construction class. If $|\chi_\nu^{\text{eng}}| \gtrsim 10^{-22}$ and Γ_{eff} is reasonably large, compact demonstrators are in the ordinary resonator and high-strength-substrate regime. If $|\chi_\nu^{\text{eng}}| \sim 10^{-23}$ to 10^{-24}, the design must move toward high-Q crystals, larger active area, static strain, or partial lift. If $|\chi_\nu^{\text{eng}}| \lesssim 10^{-25}$, compact force-device construction stops being the right first target. The useful output is a bound on the continuation.

Second, the static-versus-driven distinction is decisive. The driven column pays $uV_{\text{mode}}\omega/Q$. Static elastic storage pays no oscillator maintenance cost. It must survive creep and fracture. The same χ_ν^{eng} value can be impractical for a driven phonon design and practical for a static strain design. The static-stress experiment decides which maintenance law applies to the construction table.

$\lvert\chi_\nu^{\mathrm{eng}}\rvert$	u_{req} at $f\Sigma_2/\Gamma_{\mathrm{eff}} = 1$	P_{loss} at $f\Sigma_2/\Gamma_{\mathrm{eff}} = 1, Q = 10^5$	Construction class	Interpretation
10^{-20}	$7.3 \times 10^2\,\mathrm{J\,m^{-3}}$	$1.8\,\mathrm{W}$	small bench article	The force is easy to produce if the coherence scalar is real. Existing ambient-coherence nulls become the primary consistency check.
10^{-21}	$7.3 \times 10^3\,\mathrm{J\,m^{-3}}$	$18\,\mathrm{W}$	tabletop module	Accessible to modest driven resonators or static strain specimens. A direct balance or pendulum experiment should see it cleanly.
10^{-22}	$7.3 \times 10^4\,\mathrm{J\,m^{-3}}$	$183\,\mathrm{W}$	high-Q insert / compact platform	The first useful construction band. Full response is compatible with ordinary high-Q crystal or metal inserts. Thermal management matters.
10^{-23}	$7.3 \times 10^5\,\mathrm{J\,m^{-3}}$	$1.8\,\mathrm{kW}$	engineered crystal or metal module	Inside high-strength material energy densities. Continuous driven operation becomes hard. Static elastic storage becomes much more attractive.
10^{-24}	$7.3 \times 10^6\,\mathrm{J\,m^{-3}}$	$18\,\mathrm{kW}$	industrial multi-module system	Near the upper end of practical stored-density claims. Driven operation needs large area, high Q, cryogenic or exceptional materials, or partial-lift use.
10^{-25}	$7.3 \times 10^7\,\mathrm{J\,m^{-3}}$	$183\,\mathrm{kW}$	large array / static-only candidate	Inside the response window. Compact driven construction is implausible. Static prestress, very high Γ_{eff}, or large arrays are the plausible routes.
10^{-26}	$7.3 \times 10^8\,\mathrm{J\,m^{-3}}$	$1.8\,\mathrm{MW}$	outside compact construction	The response inequalities are satisfied. Material and power requirements exceed the intended compact-device scale. For compact devices, this row is effectively a null.

Table 4: Construction regimes inside the response-window theorem for $\Sigma_2 = 1$, $f = 1$, $u_0 = 8.5 \times 10^{-10}\,\mathrm{J\,m^{-3}}$, $V_{\mathrm{mode}} = 10^{-3}\,\mathrm{m^3}$, $\omega = 2\pi(4.0 \times 10^4)\,\mathrm{s^{-1}}$, and $Q = 10^5$. Every row has $\Delta\nu = 8.5525 \times 10^{-9}$, so construction difficulty changes while algebraic admissibility is the same. Multiply the listed material and power requirements by $f\Sigma_2/\Gamma_{\mathrm{eff}}$.

13 Experimental Bound Formula

Let an experiment have force sensitivity $F_{\min}$, projected area $A_\perp$, and known engineering-coherence contrast $\Delta S^{\mathrm{eng}}_{\mathrm{coh}}$. A null result gives

$$\boxed{|\chi^{\mathrm{eng}}_\nu| \le \frac{4\pi G F_{\min}}{g^2 A_\perp |\Delta S^{\mathrm{eng}}_{\mathrm{coh}}|}.} \tag{13.1}$$

If the mode is a driven oscillator, with $\Delta S^{\mathrm{eng}}_{\mathrm{coh}} = \Gamma_{\mathrm{eff}} Q P_{\mathrm{in}}/(u_0 \omega V_{\mathrm{mode}})$, then

$$\boxed{|\chi^{\mathrm{eng}}_\nu| \le \frac{4\pi G F_{\min} u_0 \omega V_{\mathrm{mode}}}{g^2 A_\perp \Gamma_{\mathrm{eff}} Q P_{\mathrm{in}}}.} \tag{13.2}$$

If the mode is static elastic storage, replace the driven density by u_{elastic} from Eq. (11.3).

14 Compatibility with Existing Nulls

Proposition 14.1 (No ordinary-matter anomaly from null coherence)**.** *If ordinary non-engineered configurations have $\mathfrak{c}_U = 0$ on the observer-facing scalar of Eq. (5.5), then they impose no direct bound on χ^{eng}_ν, because $\delta\nu = \chi^{\mathrm{eng}}_\nu S^{\mathrm{eng}}_{\mathrm{coh}} = 0$ for those configurations.*

Proof. Equation (5.8) gives the statement immediately. The response law couples only to the OPH-native coherent scalar. Undifferentiated rest mass or heat is outside this ansatz. □

If ordinary matter has a small nonzero ambient scalar $S^{\mathrm{eng}}_{\mathrm{coh\,amb}}$, then existing gravitational null tests bound the product:

$$|\chi^{\mathrm{eng}}_\nu|\, S^{\mathrm{eng}}_{\mathrm{coh\,amb}} \le \delta\nu_{\mathrm{test}}. \tag{14.1}$$

The condition is real. It becomes a theorem-grade number when the ambient scalar is specified. The scalar branch is falsifiable in two ways: by direct χ^{eng}_ν searches through Eq. (13.1) and by ordinary-gravity null tests once a model for $S^{\mathrm{eng}}_{\mathrm{coh\,amb}}$ is fixed.

15 Source-Lift Falsifiers

The finite source-generator branch fails if any of the following tests is confirmed under the declared receipt discipline:

1. a record-shuffled dummy with the same stored energy gives the same force signal as the active coherent material branch;
2. the inferred source changes under port relabeling, hidden carrier re-presentation, mesh relabeling, or accepted repair-schedule changes;
3. the source requires a bulk, screen-area, noncentral collar, heat, rest mass, or electrical-power scalar not carried by the edge-center scalar-slot register;
4. $S^{\mathrm{can}}_{\mathrm{coh}} = 0$ on the self-read, record, prediction, and mismatch receipts while the force is nonzero;

5. the sign of the force fails to follow the signed bottom-minus-top $\Delta S_{\rm coh}^{\rm can}$ contrast after orientation reversal and sham controls.

The dark-sector simulation planner can audit downstream galaxy, parent, Boltzmann, anomaly, and likelihood gates, but it cannot close any item in this source-lift falsifier list. In particular, a passed planner receipt does not establish `EDGE_SOURCE_LIFT_RECEIPT`, `SOURCE_NONDEGENERACY_RECEIPT`, or `VERTICAL_CONTRAST_RECEIPT`.

16 How to Read the Two Bands

The canonical band in Eq. (8.5) is the quotient-edge continuation statement. It is dimensionless and independent of material energy density. It belongs to the branch where coherent scalar sourcing is co-registered with the dark-sector scalar repair channel. On the exact uniform product-thickening branch, the value is the single number $0.9343006394893864\ldots$.

The engineering band in Eq. (8.7) is the same statement in a stored-energy chart. It depends on the declared normalization $N_{\rm coh} = \varepsilon_{\rm sub} u_{\rm stored}/u_0$. Large coherent stored energy densities make $\chi_\nu^{\rm eng}$ numerically small because the scalar itself has been scaled up.

The response window in Eq. (10.3) is an operating condition for a chosen device geometry and linear-response allowance. A design with fixed Σ, f, u, and $\Gamma_{\rm eff}$ is compatible with the OPH continuation coefficient when the forced engineering band intersects that operating window. If the forced band sits above the chosen linear ceiling, the design is outside that linear chart. If it sits below the target threshold, the target response is too large for that design.

17 Chart Choices

The canonical scalar uses the exponent-free observer core $\mathbf{1}_{\text{self-read}} R_U P_U C_U$. Optional empirical exponents or refinement factors are calibrated-chart data. They are outside the canonical quotient-edge coefficient.

The coarse-grained partition weights $w_U(x)$ should be induced by a declared observer-supporting subcover. Only the contrast entering the force law is used in the response bounds.

The exact value $e^{-P/24}$ uses the uniform product-thickening branch, including slice-wise scalar-reserve unbiasedness. The wider numerical interval in Eq. (8.5) uses finite-thickness averaging and the scalar-weighted protected reserve mean. The profile-envelope range in the remark after the exact-value corollary needs an extra numerical selector.

18 Conclusion

This note separates channel identity, coefficient value, chart conversion, and device evidence. Coherent-matter same-channel forcing gives a theorem-grade scalar-channel identity for nondegenerate OPH-coherent material sources under Scalar Edge-Center Exhaustion. The canonical quotient-edge chart and the engineering stored-energy chart are

$$\nu_{\rm eff} = \nu_{\rm OPH} + \chi_\nu^{\rm can} S_{\rm coh}^{\rm can} = \nu_{\rm OPH} + \chi_\nu^{\rm eng} S_{\rm coh}^{\rm eng}.$$

On the quotient-edge co-registered collar branch, the protected-reserve lemmas force the canonical coefficient to equal the collar survival coefficient:

$$\chi_\nu^{\rm can} = \lambda_{\rm collar}.$$

L1 to L6 give the theorem-grade band

$$0.9343006394893864\ldots \le \chi_\nu^{\rm can} \le 1.$$

L7 gives the exact value

$$\chi_\nu^{\rm can} = e^{-P/24} = 0.9343006394893864\ldots.$$

This excludes $\chi_\nu^{\rm can} = 0$ for nondegenerate coherent-material sources on the closed quotient-edge collar branch.

In the engineering chart,

$$\chi_\nu^{\rm eng} = \frac{\chi_\nu^{\rm can}}{N_{\rm coh}}, \qquad N_{\rm coh} = \varepsilon_{\rm sub}\frac{u_{\rm stored}}{u_0}.$$

This gives

$$\frac{0.9343006394893864}{N_{\rm coh}} \le \chi_\nu^{\rm eng} \le \frac{1}{N_{\rm coh}}.$$

Engineering values near 10^{-22} correspond to $N_{\rm coh} \sim 10^{22}$. The response-window inequalities then decide which device geometries and material states stay inside the chosen linear-response chart.

For $\Sigma = 100\,\mathrm{kg\,m^{-2}}$, full weight response requires

$$\Delta S^{\rm can}_{\rm coh,req} \simeq 8.6 \times 10^{-9} \text{ to } 9.2 \times 10^{-9},$$

and ten percent assist requires about 8.6×10^{-10} to 9.2×10^{-10}. A hoverboard-class areal load around 200 to $600\,\mathrm{kg\,m^{-2}}$ moves the full-response target to about 2×10^{-8} to 6×10^{-8}. The coefficient is compatible with those room-scale response levels on the closed branch. A device claim requires a substrate that produces the required signed vertical canonical contrast, plus force controls that separate acoustic, thermal, electromagnetic, and mechanical couplings. A cosmological use requires a finite covariant dark-stress parent with refinement convergence and likelihood-facing stress observables.

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